MILITARY SPECIFICATION

KEYBOARD, DATA ENTRY

This specification, is approved for use by the US Army Armament, Munitions and Chemical Command-and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Data Entry Keyboard: 9355810, used in the Mortar Ballistics Computer (MBC): M23. Throughout this specification this assembly is referenced as “keyboard”.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center. Attn: AMSMC-QA, Dover, New Jersey 07601-5001 by using the self-addressed Stamdardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Electronic Equipment, General Requirements for
MIL-STD-1521	-	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	-	Calibration- System Requirements

(Copies of Specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9355810	Keyboard, Data Entry
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9355810	Packaging of Keyboard, Data Entry: 9355810

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The Data Entry Keyboard is an assembly which provides the operator the machine interface necessary to control the operation of the Mortar Ballistics Computer (MBC) including provisions to enter and manipulate message data. The keyboard assembly contains an initialization group consisting of 11 keys, a 25 key input and edit group, a 9 key fire mission group, and a 3 key output group. The keyboard assembly also provides 4 light emitting diodes (LEDs) that indicate 4 internal states of the MBC and an electroluminescent backlight for nighttime operation. The connector contacts are listed in Table I.

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TABLE I. Keyboard assembly connector contacts.

Connector	Pin	Type	Name
P1	A1	0	HB0*
P1	A2	0	HB1*
P1	A3	0	HB2*
P1	A4	0	HB3*
P1	A5	0	HB4*
P1	A6	0	HB5*
P1	A7	0	HB6*
P1	A8	N/A	GND
P1	A9	I	MSGLED
P1	A10	I	ONLED
P1	A11	N/A	ELLO
P1	B1	0	LBO*
P1	B2	0	LB1*
P1	B4	0	LB3*
P1	B5	0	LB4*
P1	B6	0	LB5*
P1	B7	0	LB6*
P1	B8	0	ON*
P1	B9	I	SEQLED
P1	B10	I	BATLED
P1	B11	N/A	ELH1

3.2 Performance.

3.2.1 General. The performance of the keyboard shall be in accordance with the following requirements.

3.2.1.1 Key closure. When the following keys are pressed with a force of 16 plus or minus 8 ounces, there shall be continuity between the associated connector contacts and P1-A8 (GND) and the resistance shall be less than 7.5 ohms.

Key	Connector Contacts	
SET UP	P1-A1	P1-B1
KNPT/TGT	P1-A1	PA-B1
EOM	P1-A3	P1-B1
3 GHI	P1-A4	P1-B1
MSG	P1-A5	P1-B1
XMIT	P1-A6	P1-Bl
REG	P1-A7	P1-B1
WPN DATA	P1-A1	P1-B2
MET	P1-A2	P1-B2
COMPUTE	P1-A3	P1-B2
4 JKL	P1-A4	P1-B2
REVIEW	P1-A5	P1-B2
GRID	P1-A6	P1-B2

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TFC	P1-A7	P1-B2	
FO LOGIC	P1-A1	P1-B3	
FIRE ZONES	P1-A2	P1-B3	
CLEAR ENTRY	P1-A3	P1-B3	
5 MNO	P1-A4	P1-B3	
SURV	P1-A5	P1-B3	
SHIFT	P1-A6	P1-B3	
FPF	P1-A7	P1-B3	
REG DATA	P1-A1	P1-B4	
DSPI	P1-A2	P1-B4	
-	P1-A3	P1-B4	
6 PQR	P1-A4	P1-B4	
MSN	P1-A5	P1-B4	
POLAR	P1-A6	P1-B4	
FIRE DATA	P1-A7	P1-B4	
BRT	P1-A1	P1-B5	
DSP2	P1-A2	P1-B5	
0	P1-A3	P1-B5	
7 STU	P1-A4	P1-B5	
SEQ	P1-A5	P1-B5	
ADJ	Pl-A6	P1-B5	
SFTY DATA	P1-A7	P1-B5	
TEST	P1-A1	P1-B6	
DSP 3	P1-A2	P1-B6	
1 ABC	P1-A3	P1-B6	
8 VWX	P1-A4	P1-B6	
BACK	P1-A5	P1-B6	
WPN/AMMO	P1-A6	P1-B6	
REPLOT	P1-A7	P1-B6	
AMMO DATA	P1-A1	P1-B7	
DSP4	P1-A2	P1-B7	
2 DEF	P1-A3	P1-B7	
9 YZ	P1-A4	P1-B7	
ON/OFF	P1-A5	P1-B7	P1-B8
BURST	P1-A6	P1-B7	

3.2.1.2 Light emitting diode (LEDs) illumination. The (LEDs) shall be clearly visible when observed at light levels of zero to ten foot-candles at a distance of 18 to 20 inches from the viewers keyboard.

3.2.1.3 Electroluminescent backlight (ELB) illumination. The ELB shall illuminate all keys so that the keys are clearly visible when observed at light levels of zero to ten foot-candles at a distance of 18 to 20 inches from the viewers eye when a 35 plus or minus 1 volt rms sine wave is connected to the keyboard.

3.2.2 Physical characteristics. The physical characteristics shall be as prescribed by the drawings, specifications, and QAP's.

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3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques employed in the fabrication of the keyboard shall not degrade the inherent reliability. The inherent reliability shall not be less than 162,056 hours Mean Time Between Failures (MTBF). (See 4 and 6)

3.3 Environmental. The keyboard shall meet the requirements specified herein unless otherwise specified.

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the keyboard thermally stabilized at room ambient temperature of +16°C to +32°C (+60°F to +90°F), unless otherwise specified in the requirements paragraph.

3.3.2 Environmental stress screen (ESS) temperature. The keyboard shall meet the requirements of 3.2.1.1 and 3.2.1.3 inclusive after having been exposed to a minimum of ten temperature cycles at temperatures of -57°C and +71°C (-71°F and +160°F).

3.3.3 Environmental stress screen (ESS) vibration. The keyboard shall meet the requirements of 3.2.1.1 and 3.2.1.3 inclusive, after having been subjected to a random or pseudorandom vibration-which includes frequency content from 100 Hz to 1000 Hz and up to a 3g RMS in two axis for a period of 10 minutes plus or minus 15 seconds.

3.4 Fabrication. The keyboard shall be manufactured in accordance with drawing 9355810.

3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electrical connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) keyboards plus three (3) each of all items covered by QAP. The three keyboards shall be tested as specified herein for all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of their contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractor's inspections.

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4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with “Submission of Product” provision of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP’s) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4 Sampling.

4.4.1 Environmental/performance. The keyboard shall be inspected to the requirements in Table II.

TABLE II. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
100% Inspection			
1.	Key closure	3.2.1.1	4.6.3
2.	LED illumination	3.2.1.2	4.6.4
3.	ELB illumination	3.2.1.3	4.6.5
4.	ESS temperature	3.3.2	4.6.1
5.	ESS vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective			
101.	Workmanship	3.6	4.6.6

4.4.2 Special sampling. One display\processor circuit board. assembly shall be selected at random as a special sample from each. lot or each 100 produced which ever occurs first. The sample Shall meet the requirements and tests in Table III.

TABLE III. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing\Visual
202.	Materials	3.7	Applicable Drawing\Visual

NOTE: The tests in Table III shall be conducted at room anbient temperature (as specified in 3.3.1).

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4.4.3 Failure of sample. Should any one item of a special sampling fall to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate. acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed

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0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of he set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The keyboard shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The keyboard shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 10°c (18°F) per minute plus or minus 1°C (1.8°C) to +71°F (+160°C). The keyboard shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) to room ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. Upon completion of the cycles the keyboard shall then be removed from the chamber and subjected to the tests in 4.6.3 and 4.6.5 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.3 inclusive.

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4.6.2 ESS vibration. Conduct the vibration test with the keyboard positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The applied motion shall be applied in two separate planes for the time period specified. Upon completion of the vibration sweep the keyboard shall be removed from the vibration adapter and shall be subjected to the tests in 4.6.4 and 4.6.5 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.3 inclusive.

4.6.3 Key closure. This test is to be conducted utilizing an ohm meter and appropriate spring scale. With the ohm meter attached to the key to be tested and P1-A8 (GND), verify continuity and measure (with the spring scale) the force required to reduce resistance to less than 7.5 ohms. The force, resistance, and continuity shall be as specified in 3.2.1.1.

4.6.4 Light emitting diodes (LEDs). Conduct this test by connecting one end of a 200 ohm plus or minus 5 percent 1\4 W resistor (4 total) to each of the 4 LED contacts listed in Table I. Connect the other end of the resistors to a 5.3 plus or minus 0.1 void DC power supply that is referenced to GND. The output response shall be as specified in 3.2.1.2.

4.6.5 Electroluminescent backlight (ELB). Conduct this test by connecting a 35 plus or minus 1 volt rms sine wave generator to P1-B11 with respect to P1-A11. The output response shall be as specified in 3.2.1.3.

4.6.6 Workmanship. The keyboard shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The assembly shall conform to the requirements of 3.6.

4.7 Reliability testing. The requiremnt for reliability test approval and the responsibility (Government or contractor) for reliability testing shall be as specified in the contract. The sample for reliability test shall consist of 3 keyboards. The samples shall be manufactured in the same manner, using the same materials equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

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4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and tested by the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELIVERY

5.1 Delivery requirements. Handling, packaging, packing, shipping, and marking of the Data Entry Keyboard, 9355810 shall be in accordance with Packaging Data Sheet 9355810. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The keyboard described herein is intended for use in the MBC. See drawing 11785850.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

e. Applicable levels of preservation, packing, and marking (see 501).

f. Applicable stock numbers.

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MIL-K-70493 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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